3 **Correlation**	**3.1**	**Use of formulae to calculate the product moment correlation coefficient.** **Knowledge of the conditions for the use of the product moment correlation coefficient.** **A knowledge of the effects of coding will be expected.**	**Students will be expected to be able to use the formula to calculate the value of a coefficient given summary statistics.**
	3.2	**Spearman's rank correlation coefficient, its use, interpretation.**	**Use of** $r_s = 1 - \frac{6\sum d^2}{n(n^2-1)}$ **Numerical questions involving ties may be set. An understanding of how to deal with ties will be expected.** **Students will be expected to calculate the resulting correlation coefficient on their calculator or using the formula.**

Topics	What students need to learn:		
	Content		Guidance
3 **Correlation** ***continued***	3.3	**Testing the hypothesis that a correlation is zero using either Spearman's rank correlation or the product moment correlation coefficient.**	**Hypotheses should be in terms of ρ or ρ_s and test a null hypothesis that ρ or $\rho_s = 0$.** **Use of tables for critical values of Spearman's and product moment correlation coefficients.** **Students will be expected to know that the critical values for the product moment correlation coefficient require that the data comes from a population having a bivariate normal distribution. Formal verification of this condition is not required.**